Vaccines and How Immunity Works

Introduction to Immunity

Immunity is the body's ability to recognise and defend itself against harmful germs, called pathogens, such as bacteria, viruses, fungi and parasites. The body's defence is carried out by the immune system, a network of cells, tissues and organs including white blood cells, bone marrow, the thymus, the spleen, lymph nodes and the skin. Innate immunity is the first line of defence, which acts quickly and in the same way against all germs; it includes skin, mucus, stomach acid, fever, inflammation and cells such as macrophages and neutrophils. Adaptive immunity is slower to start but specific and has memory; it uses B cells, which make proteins called antibodies, and T cells, which destroy infected cells. After the body has met a germ once, memory cells remain and can respond faster and stronger the next time. This memory is the principle behind vaccines.

How Does a Vaccine Train the Immune System?

A vaccine is a biological preparation that trains the immune system to recognise a disease-causing germ without causing the disease itself. It contains a harmless form or piece of the pathogen, called an antigen, or instructions for making one. When the vaccine is injected or given by mouth, the immune system identifies the antigen as foreign and responds by producing antibodies and activating T cells. Memory B and T cells are created. If the person later encounters the real germ, these memory cells recognise it immediately and launch a rapid, strong response that stops the disease or makes it much milder. In this way, a vaccine gives "practice" to the body at little risk. Some vaccines need booster doses to maintain protection. Vaccination is one of the most successful public health measures, saving an estimated 3 to 5 million lives every year according to the World Health Organization.

Types of Vaccines

Live attenuated vaccines use a weakened form of the germ, as in the measles, mumps and rubella (MMR) vaccine, the oral polio vaccine, and the BCG vaccine for tuberculosis. Inactivated vaccines use killed germs, as in the injectable polio vaccine and hepatitis A. Subunit, recombinant and conjugate vaccines use only a part of the germ, such as the hepatitis B and HPV vaccines. Toxoid vaccines use inactivated toxins, as against tetanus and diphtheria. Viral vector vaccines use a harmless virus to carry genetic instructions; the Oxford-AstraZeneca vaccine, sold in India as Covishield, is an example. mRNA vaccines, such as those of Pfizer-BioNTech and Moderna against COVID-19, deliver a messenger RNA that teaches cells to make a harmless piece of the virus; Katalin Kariko and Drew Weissman won the 2023 Nobel Prize in Physiology or Medicine for the research enabling them.

What Is Herd Immunity?

Herd immunity, also called community immunity, occurs when a large enough proportion of a population is immune to a disease, either through vaccination or previous infection, so that the disease cannot spread easily. This indirectly protects people who cannot be vaccinated, such as newborn babies, people with weakened immune systems and some patients. The percentage needed depends on how contagious the disease is, measured by the basic reproduction number (R0). Highly contagious measles needs about 95 percent immunity, while polio needs about 80 percent. When vaccination rates fall, outbreaks can return. Herd immunity works only when immunity is achieved safely through vaccines, since reaching it through infection causes illness and deaths.

Who Developed the First Vaccine?

The first vaccine was developed by the English physician Edward Jenner in 1796. He noticed that milkmaids who had caught cowpox, a mild disease, did not catch deadly smallpox. On 14 May 1796 he inoculated an eight-year-old boy, James Phipps, with material from a cowpox sore, and later showed that the boy was protected from smallpox. The word vaccine comes from the Latin "vacca", meaning cow. Louis Pasteur developed vaccines against chicken cholera, anthrax and, in 1885, rabies, and he extended the word vaccine to all such preparations. Smallpox was declared eradicated by the World Health Organization on 8 May 1980, the only human disease eradicated so far, with India reporting its last case in 1975. Variolation, a risky older practice of inoculating with smallpox material, was also known in India and China long before Jenner.

Vaccination in India

India runs one of the world's largest immunisation programmes. The Universal Immunisation Programme, launched in 1985, provides free vaccines against diseases such as tuberculosis, polio, diphtheria, whooping cough, tetanus, hepatitis B, measles and rotavirus diarrhoea to millions of children and pregnant women, and Mission Indradhanush, launched in 2014, targeted children who had missed vaccines. India was certified polio-free by the WHO on 27 March 2014, after the last case on 13 January 2011 in West Bengal. During the COVID-19 pandemic, India began its vaccination drive on 16 January 2021 with the homegrown Covaxin, developed by Bharat Biotech with the Indian Council of Medical Research, and Covishield, made by the Serum Institute of India in Pune, the world's largest vaccine manufacturer by number of doses. India administered over 2 billion doses through the CoWIN platform. In Tamil Nadu, the King Institute of Preventive Medicine and Research in Chennai and the Pasteur Institute of India in Coonoor have a long history of vaccine production and rabies work.

Safety and Myths

Vaccines are tested in clinical trials and monitored continuously. Mild side effects such as a sore arm, fever or tiredness are common and short-lived; serious reactions are very rare. Vaccines do not cause the diseases they prevent, and the claim that vaccines cause autism has been disproved by many large studies.

Challenges and Future

Challenges include vaccine hesitancy caused by misinformation, cold-chain storage in a hot country and new variants. Future vaccines include those for malaria, dengue, tuberculosis and cancer, and needle-free forms.

Quick Facts

- A vaccine trains the immune system with a harmless antigen so that it makes antibodies and memory cells.
- Herd immunity protects a community when enough people are immune, which is about 95 percent for measles.
- Edward Jenner developed the first vaccine, for smallpox, in 1796.
- India was certified polio-free on 27 March 2014.
- India's COVID-19 vaccination drive began on 16 January 2021 with Covaxin and Covishield.